Q1. (IA Groups ES1/Q1) Let G be any group. Show that the identity e is the only element $g \in G$ satisfying the equation $g^2 = g$.

Q2. (IA Groups ES1/Q4) Let G be a finite group.

(a) Show from first principles that there exists an $n \in \mathbb{Z}^+$ such that $g^n = e$.

(b) Show from first principles that there exists an $n \in \mathbb{Z}^+$ such that $g^N = e$ for all $g \in G$.

Q3. (IA Groups ES1/Q3) Let $G = \mathbb{R} \setminus \{-1\}$, and define a binary operation $\circ$ on G:

$$x \circ y = x + y + xy$$

(a) Show that $(G, \circ)$ is a group.

(b) What is the inverse of 2?

(c) Solve the equation $2 \circ x \circ 5 = 6$

Q4. Let G be a finite group of order 28 and consider its subgroups $H \leq G$ and $K \leq G$ with $|H| = 4$ and $|K| = 7$. There exists an element $g \in G$ such that $g \notin H \cup K$. Show that either $|g| = 14$ or $|g| = 28$.

Q5. Let H and K be subgroups of the group G.

(a) (IA Groups ES1/Q2) Prove that $H \cap K$ is also a subgroup of G.

(b) Suppose $|H| = p$ and $|K| = q$ where p, q are distinct primes. What is the order of the subgroup $H \cap K$?

Q6. (IA Groups ES1/Q11) Let G be a group in which every element, other than the identity, has order two. Show that G is abelian.

Q7. Let G be the set of all matrices of the form $\begin{pmatrix} a & b \\ -b & a \end{pmatrix}$ where $a, b \in \mathbb{R}$ and $a^2 + b^2 \neq 0$, under standard matrix multiplication. State a group isomorphic to G.

Q8. (IA Groups ES1/Q12) Let G be a finite group of even order.

(a) Show that G contains an element of order two.

(b) Can a group have exactly two elements of order two? If so, find a group which has this property. If not, explain why.